

0 6 . 1

Each year, a few people with type I diabetes are given a pancreas transplant. Pancreas transplants are not used to treat people with type II diabetes.

Give **two** reasons why pancreas transplants are not used for the treatment of type II diabetes.

[2 marks]

1 _____

2 _____

0 6 . 2

The pancreas produces the hormone insulin.

Put a tick (✓) in the box next to the statement which describes **incorrectly** the action of insulin.

[1 mark]

Activates enzymes involved in the conversion of glucose to glycogen.

☐

Controls the uptake of glucose by regulating the inclusion of channel proteins in the surface membranes of target cells.

☐

Attaches to receptors on the surfaces of target cells.

☐

Activates enzymes involved in the conversion of glycerol to glucose.

☐

Question 6 continues on the next page

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Scientists investigated the use of induced pluripotent stem cells (iPS cells) to treat type I diabetes in mice. The scientists used four transcription factors to reprogramme skin cells to form iPS cells. The scientists then stimulated the *in vitro* differentiation of iPS cells into pancreatic cells.

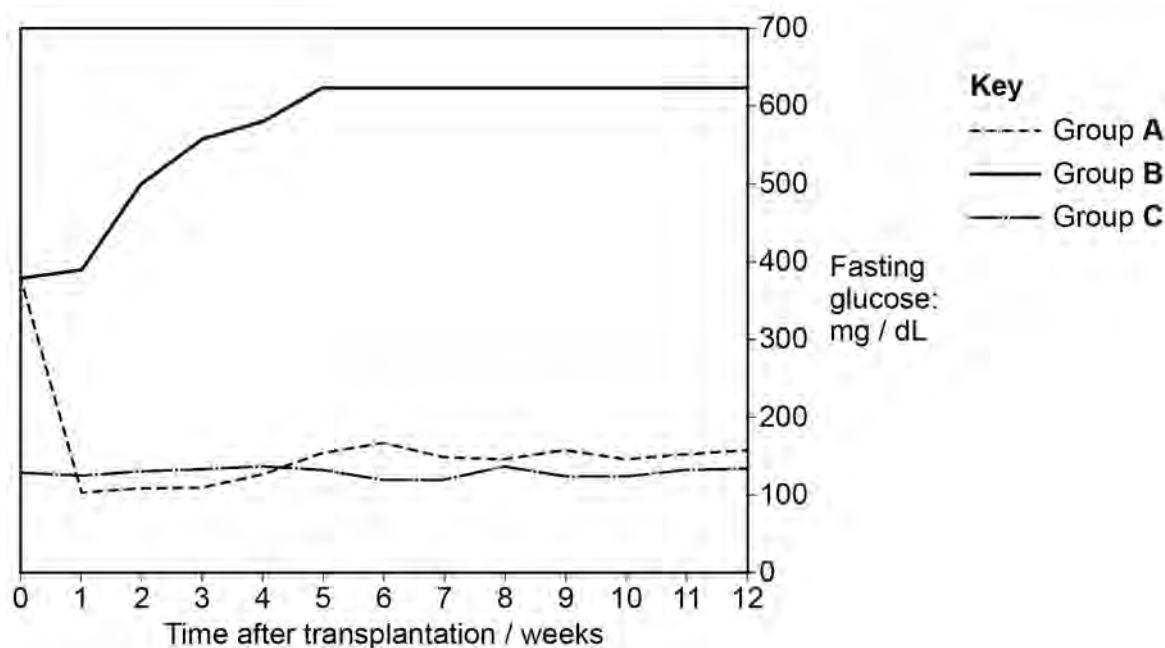
The scientists set up three experimental groups:

- Group **A** – 30 mice with type I diabetes received pancreatic cell transplants derived from iPS cells.
- Group **B** – 30 mice with type I diabetes were left untreated.
- Group **C** – 30 mice without diabetes were left untreated.

The scientists measured the blood glucose concentration of all the mice on a weekly basis for 12 weeks.

The results the scientists obtained are shown in **Figure 6**.

Figure 6



Suggest how transcription factors can **reprogramme** cells to form iPS cells.

[2 marks]

Using all the information provided, evaluate the use of iPS cells to treat type I diabetes in humans.

[illegible]

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